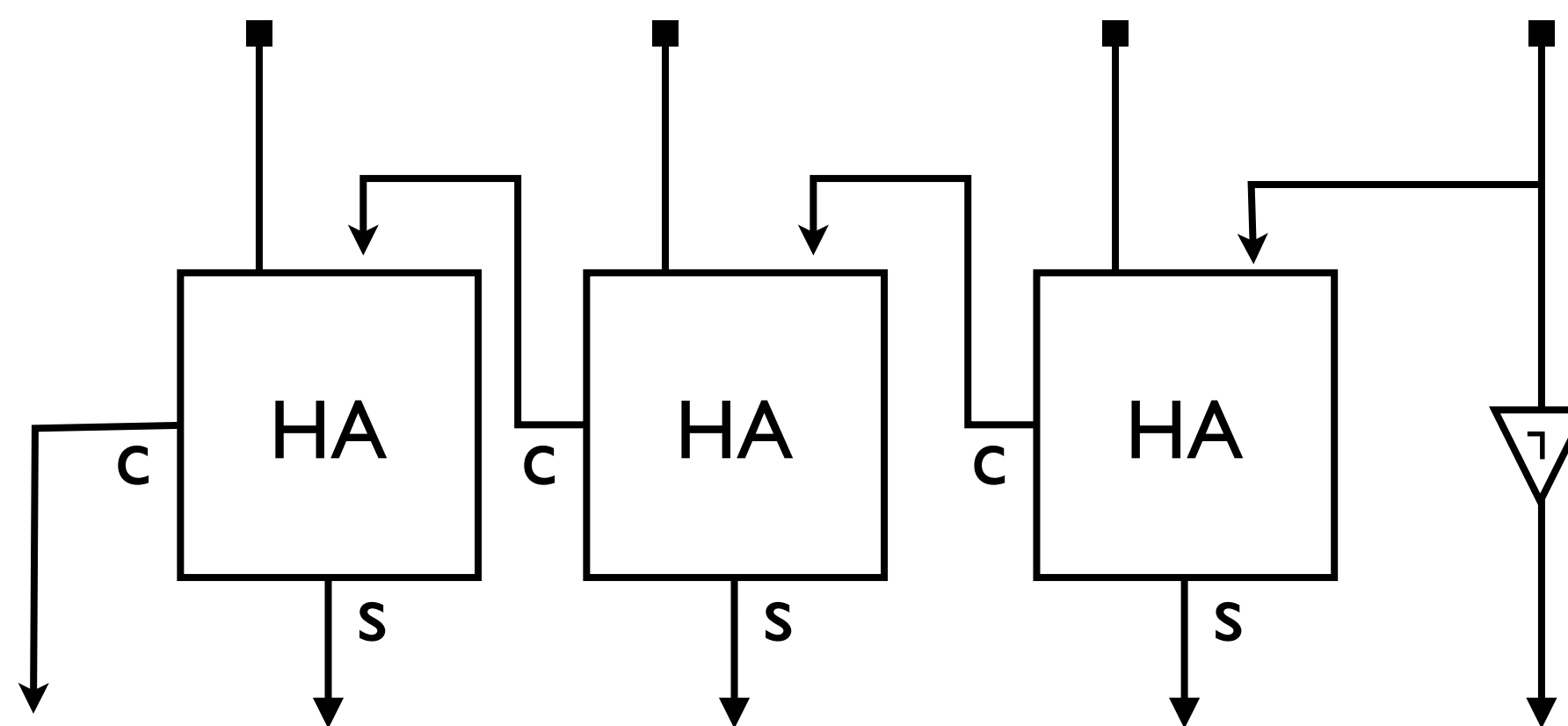


10. Lessons from Binary Successor

Fundamental lessons about the nature of computation



+1[0-15] B combinatorial circuit

Lesson 1: The physical realization of a cognition.

+1B shows us definitively that an entirely physical system can perform cognitive operations, via the method of symbolic simulation.

Lesson 2: Information processing as functional description.

Hammers have the function of pounding nails, and hearts have the function of pumping blood. +1B has an *information processing* function, which goes beyond the way it manipulates the physical world. “It computes +1” is the informational-level description of the machine. Compare +1B and +1T: different physical functions, same information processing function.

Lesson 3: Computing a function vs implementing a function.

All things *implement* the laws of physics, but only some things *compute* the laws of physics. To compute is to *represent* a function, but not to actually conform with it. +1B can compute the laws of *adding-apples-to-a-basket*, but it does not *implement* these laws. (By the way, do the planets compute their own orbit? When humans walk, do they compute or implement their trajectory?)

Lesson 4: Computation does not require an explicitly represented algorithm.

+1B computes a function not by “following” a set of rules, which would require an inner-agent to do the following, but simply by *embodying* a set of rules in its functional architecture. To compute a function is to

Lesson 5: Competence vs performance.

+1B shows how a physical system can have certain cognitive *capacities*— i.e. the +1B machine has the capacity to compute +1, for a range of arguments, even if it is never actually used for this. It also illustrates the contrast between the capacities you have *competence* for, and your actual performance. Because of entropy, friction, and the rest, how a machine actually performs can pull apart from its competence.